ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 3RD YEAR

COMPUTER NETWORKS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 3rd year Computer Networks paper 1 · practice a.

Written by Academic Central for practice, following third-year Computer Science syllabuses at South African universities (computer networks, top-down approach). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct working even if a later step is wrong (consistent accuracy). Accept other correct protocols, examples and wording.

QUESTION 1

1.1 B ✓
1.2 D ✓
1.3 C ✓
1.4 C ✓
1.5 B ✓
1.6 A ✓
1.7 B ✓
1.8 B ✓
1.9 C ✓
1.10 A ✓

(10)

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QUESTION 2

2.1 Application layer: HTTP, DNS or SMTP ✓ (5)
Transport layer: TCP or UDP ✓
Network layer: IP or ICMP ✓
Link layer: Ethernet or Wi-Fi (802.11) ✓
Physical layer: twisted-pair copper, fibre or radio ✓
(Layer and example both needed for each mark.)

2.2 Each layer takes the data from the layer above and adds its own header (and the link layer also a trailer) ✓, so a message becomes a segment, then a datagram, then a frame ✓ (2)

2.3 Circuit switching reserves a dedicated path and fixed bandwidth for the whole call ✓; packet switching splits data into packets that share the links and are sent as they come, without reservation ✓ (4)
Advantage of circuit switching: guaranteed, steady rate and delay (good for voice) ✓
Advantage of packet switching: links are used efficiently, so many more users can share them ✓

2.4.1 $L = 1\,500 \times 8 = 12\,000$ bits ✓ (2)
$d_{trans} = L / R = 12\,000 / (10 \times 10^6) = 0.0012$ s $= 1.2$ ms ✓

2.4.2 $d_{prop} = d / s = 2 \times 10^6$ m $/ 2 \times 10^8$ m/s $= 0.01$ s $= 10$ ms ✓ (1)

2.4.3 1.2 + 10 = 11.2 ms ✓ (1)

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QUESTION 3

3.1.1 The data has four 1s, so the parity bit is 0 ✓ (1)

3.1.2 Flipping two bits changes the number of 1s by 0 or by 2 ✓, so the count stays even and the parity check still passes; the error is not seen ✓ (2)

3.2 Append three 0s (degree of G is 3): 1101011**000** ✓ (5)
Divide by 1011 using XOR (modulo-2) ✓:
1101 ⊕ 1011 = 0110 → 1100 ⊕ 1011 = 0111 → 1111 ⊕ 1011 = 0100 → 1001 ⊕ 1011 = 0010 → 1000 ⊕ 1011 = 0011 ✓
Remainder (CRC) = 110 ✓
Frame sent: 1101011110 ✓

3.3 A node listens to the channel and only sends when it is idle (carrier sense) ✓ (4)
While sending it keeps listening; if it detects a collision it stops and sends a jam signal ✓
It then waits a random back-off time (binary exponential back-off) before trying again ✓
On switched full-duplex Ethernet each link joins one host to one switch port and can send in both directions at once, so collisions cannot happen ✓

3.4 The switch learns that A is reached through port 1 (it stores A's MAC address, port 1 and a time) ✓ (3)
B is not in the table, so the switch floods the frame out of every port except port 1 ✓
When B replies, the switch learns B's port and from then on forwards frames to B only on that port ✓

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QUESTION 4

4.1.1 4 subnets need 2 extra bits: /26 ✓ 255.255.255.192 ✓ (2)

4.1.2 (8)

Subnet	Network address	Usable hosts	Broadcast address
1	196.21.40.0	196.21.40.1 – 196.21.40.62	196.21.40.63
2	196.21.40.64	196.21.40.65 – 196.21.40.126	196.21.40.127
3	196.21.40.128	196.21.40.129 – 196.21.40.190	196.21.40.191
4	196.21.40.192	196.21.40.193 – 196.21.40.254	196.21.40.255

Two marks per subnet: network and broadcast address ✓ host range ✓

4.1.3 $2^6 - 2 = 62$ ✓ (1)

4.2 (a) Interface 2 (the /24 is the longest match) ✓ (4)
(b) Interface 1 (10.1.4.0/22 covers 10.1.4.0 to 10.1.7.255) ✓
(c) Interface 0 ✓
(d) Interface 3 (default) ✓

4.3 (7)

Step	N′	D(v), p(v)	D(w), p(w)	D(x), p(x)	D(y), p(y)	D(z), p(z)
0	u	2, u	7, u	3, u	∞	∞
1	uv		6, v	3, u	∞	∞
2	uvx		4, x		6, x	∞
3	uvxw				5, w	10, w
4	uvxwy					7, y
5	uvxwyz					

Steps 0–1 ✓ step 2 ✓ step 3 ✓ steps 4–5 ✓
Least-cost paths: v 2 (u–v), x 3 (u–x), w 4 (u–x–w), y 5 (u–x–w–y), z 7 (u–x–w–y–z) ✓
Forwarding table of u: destination v → link (u, v) ✓; destinations x, w, y and z → link (u, x) ✓

4.4 In link-state routing every router learns the whole network map and works out the paths itself (Dijkstra) ✓; in distance-vector routing each router only shares its distance table with its neighbours and uses Bellman-Ford ✓ (3)
The problem is count-to-infinity ✓

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QUESTION 5

5.1 Any THREE: (3)
TCP is connection-oriented (handshake); UDP is connectionless ✓
TCP gives reliable, in-order delivery with ACKs and retransmission; UDP does not ✓
TCP has flow and congestion control; UDP has none ✓
(Also: TCP is a byte stream, UDP keeps message boundaries; UDP has a smaller 8-byte header.)

5.2 Any one: live video or voice calls, online games, DNS ✓ (2)
Speed matters more than a lost packet: a late retransmission is useless, and UDP has no handshake or congestion slow-down ✓

5.3 Client → server: SYN, seq = 100 ✓ (4)
Server → client: SYN and ACK, seq = 300, ack = 101 ✓✓
Client → server: ACK, seq = 101, ack = 301 ✓

5.4.1 100 ✓ (1)

5.4.2 92 (B is still waiting for byte 92, so it repeats that ACK) ✓ (1)

5.4.3 When the sender receives THREE duplicate ACKs for the same data ✓, it resends the missing segment at once, without waiting for its timer to run out ✓ (2)

5.5.1 Rounds 1–5: 1, 2, 4, 8, 16 (doubles every RTT in slow start) ✓✓ (3)
cwnd has reached ssthresh, so congestion avoidance adds 1 MSS per RTT: round 6 = 17, round 7 = 18 ✓

5.5.2 ssthresh = 18 / 2 = 9 MSS ✓ cwnd = 9 MSS (fast recovery, then congestion avoidance) ✓ (2)
(Accept cwnd = 12 if the three extra MSS were added.)

5.5.3 ssthresh = 9 MSS and cwnd = 1 MSS ✓; TCP goes back to slow start ✓ (2)

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QUESTION 6

6.1 The host asks its local DNS server for www.academiccentral.co.za ✓ (4)
The local DNS server asks a root server, which replies with the servers for the .za top-level domain ✓
It asks a .za server, which refers it to the co.za servers, which refer it to the authoritative server for academiccentral.co.za ✓
The authoritative server returns the IP address; the local server caches it and sends it to the host ✓

6.2.1 Non-persistent HTTP opens a new TCP connection for every object and closes it afterwards ✓; persistent HTTP keeps one connection open and fetches many objects over it ✓ (2)

6.2.2 (a) 5 objects × 2 RTT (TCP set-up + request) = 10 RTT ✓ (2)
(b) 2 RTT for the connection and the HTML + 4 × 1 RTT for the images = 6 RTT ✓

6.3 (a) SMTP ✓ (b) IMAP ✓ (2)

6.4 Symmetric key: the same secret key encrypts and decrypts; it is fast ✓ (3)
Public key: a public key encrypts (or checks signatures) and only the matching private key decrypts (or signs); no shared secret is needed beforehand, but it is slow ✓
TLS uses public-key cryptography and the server's certificate in the handshake to agree on a shared session key, then uses that symmetric key to encrypt the data ✓

6.5 A stateful firewall keeps track of open connections ✓, so it can let in only packets that belong to a connection started from inside (e.g. a TCP reply), while a packet filter looks at each packet on its own using fixed rules ✓ (2)

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TOTAL: 100

END OF MARKING GUIDELINE